A Reproducible Study of Queue Policies

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$$\dot{x} = \sigma(y - x),$$
$$\dot{y} = x(\rho - z) - y,$$
$$\dot{z} = xy - \beta z$$

σ=10, ρ=28, β=$\frac{8}{3}$

Abstract

This working-paper template separates motivation, method, results, and limitations. Its queue-policy example uses synthetic illustrative data. It establishes a reporting structure and does not report an executed experiment or a validated performance improvement.

Keywords: reproducibility; queue policies; experimental design.

1 Introduction and related work

A useful study defines its question and baseline before reporting results. Foundational texts illustrate technical exposition and precise notation [1, 2]; these references are bibliography demonstrations, not evidence for the example queue-policy comparison.

2 Research question and methodology

Question: How does a scheduling policy affect mean response time at a fixed offered load? State the workload distribution, random seeds, warm-up, replicate count, confidence interval method, and hardware/software environment.

$$\bar{t} = \frac{1}{n} \sum_{i=1}^{n} t_i. \tag{1}$$

Algorithm 1 Proposed comparison protocol

Require: Workloads W, policies P, replicate count n

for all $w \in W$ **do**
  for all $p \in P$ **do**
    for $i = 1$ to n **do**
      Run (w, p) with a recorded seed and collect response times
    end for
    Summarize uncertainty across independent replicates
  end for
end for

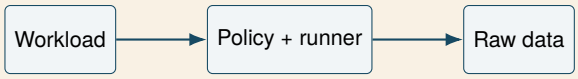


Figure 1: Proposed experimental pipeline.

3 Illustrative results

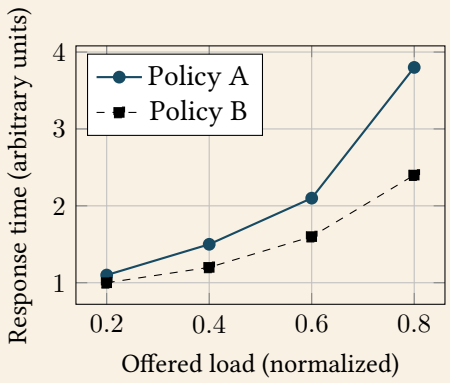


Figure 2: Synthetic values loaded from a CSV; no measurement or uncertainty claim.

Table 1: Reporting checklist for an actual study.

Item	Required record
Environment	Processor, memory, operating system, tool versions
Workload	Dataset provenance, input sizes, seeds, baseline
Statistics	Independent replicates, uncertainty, exclusions
Artifacts	Source revision, scripts, raw data, license

4 Discussion and threats to validity

Do not infer real-world superiority from figure 2. Replace the illustrative CSV with actual data and describe the collection process. Discuss internal validity, generalizability, and uncertainty before claiming an effect.

5 Conclusion and availability

Summarize only supported findings. Identify unresolved questions and provide artifact locations. For a proposal, replace results with planned evaluation and milestones. For submission, adopt the venue's official class and style.

References

[1] Donald E. Knuth. *The Art of Computer Programming: Fundamental Algorithms*, volume 1. Addison-Wesley, 3 edition, 1997.

[2] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x.